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Set 17: Employee Performance Analysis

Data Visualization using R

Aim

To analyse employee performance using a line chart, a bar chart, a table and a dashboard in R.

Procedure

1. Open RStudio and load the packages ggplot2 and gridExtra (install them once with `install.packages(c("ggplot2", "gridExtra"))`).
2. Create the employee data as a data frame.
3. Question 1: draw the line chart of performance score against years of service with `geom_line()`.
4. Question 2: draw the bar chart of employees per department with `geom_bar()`.
5. Question 3: make the table with `tableGrob()`.
6. Question 4: join the charts and the table into a dashboard with `arrangeGrob()`.
7. Run the code and note the output and the figures.

R Code

```
library(ggplot2); library(gridExtra)

# Dataset
emp <- data.frame(Employee_ID = 1:3, Department = c("Sales", "HR", "Marketing"),
                  Years = c(5, 3, 7), Score = c(85, 92, 78))
print(emp)
cat("Average score:", mean(emp$Score), "\n")

# 1. Line chart - performance over years of service
p1 <- ggplot(emp, aes(Years, Score)) + geom_line() + geom_point(size = 3) +
  geom_text(aes(label = Department), vjust = -1) +
  labs(title = "Performance vs Years of Service", x = "Years of Service", y = "Performance Score")

# 2. Bar chart - employees per department
p2 <- ggplot(emp, aes(Department, fill = Department)) + geom_bar(show.legend = FALSE) +
  labs(title = "Employees per Department", x = "Department", y = "Employees")

# 3. Table
tb <- tableGrob(emp, rows = NULL)

# 4. Dashboard
dash <- arrangeGrob(p1, p2, tb, layout_matrix = rbind(c(1,2), c(3,3)), top = "Employee Performance Dashboard")

ggsave("set17_q1.png", p1, width = 6, height = 4, bg = "white")
ggsave("set17_q2.png", p2, width = 6, height = 4, bg = "white")
ggsave("set17_q3.png", arrangeGrob(tb, top = "Employee Performance Table"), width = 5, height = 2, bg = "white")
ggsave("set17_q4.png", dash, width = 12, height = 8, bg = "white")
```

Output

```
Employee_ID Department Years Score
1          1    Sales     5     85
2          2      HR      3     92
3          3 Marketing     7     78
Average score: 85
```

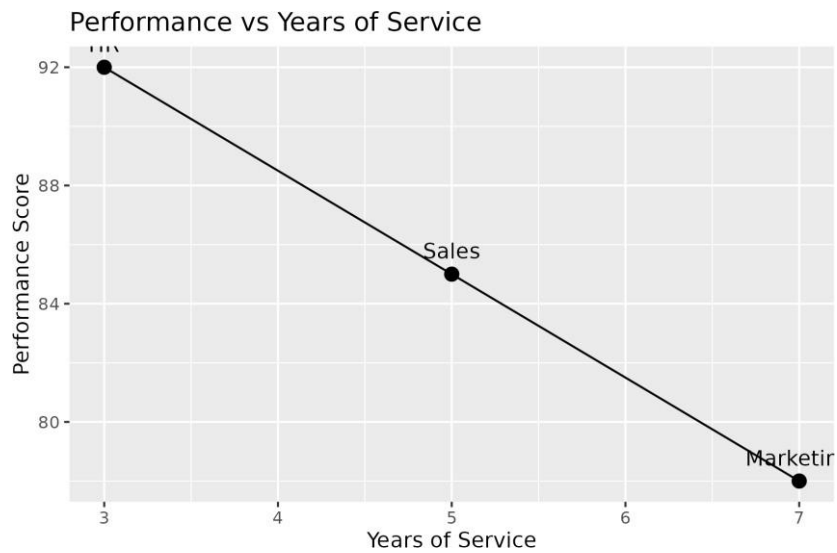


Figure 17.1: Q1 - Performance line chart

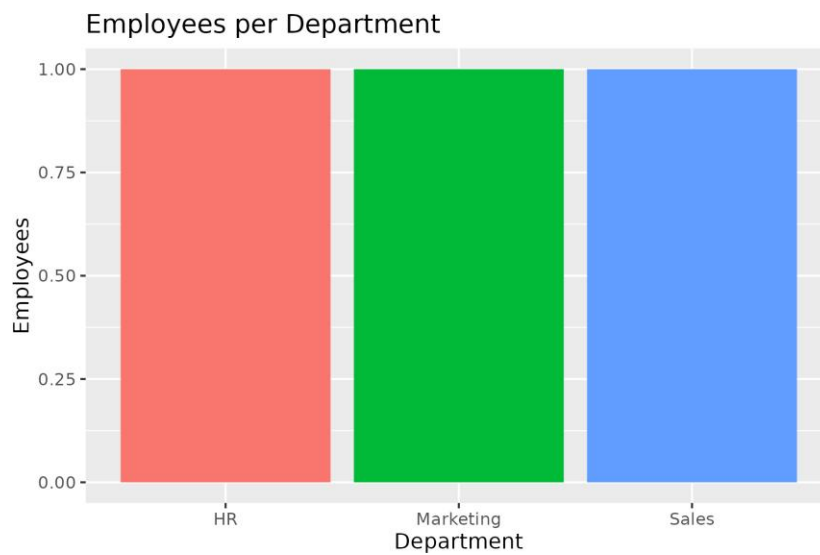
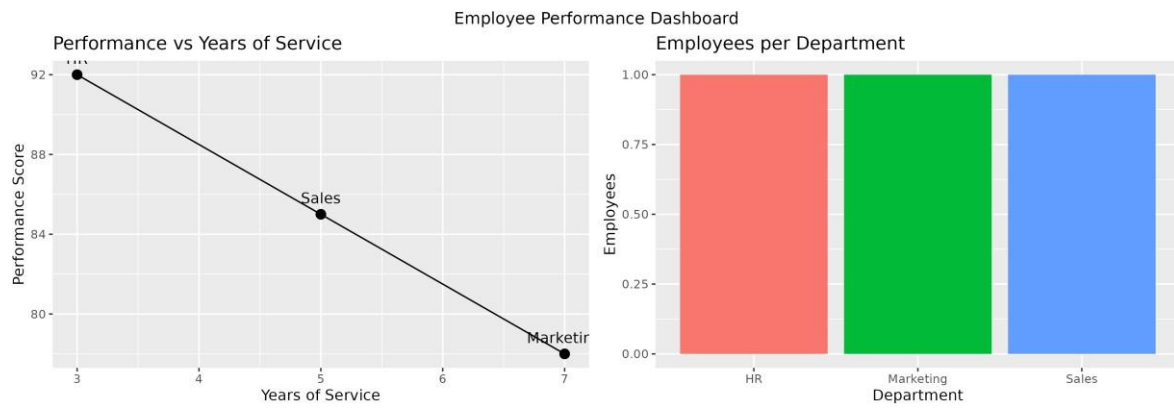


Figure 17.2: Q2 - Employees per department

Employee Performance Table

Employee_ID	Department	Years	Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

Figure 17.3: Q3 - Employee performance table



Employee_ID	Department	Years	Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

Figure 17.4: Q4 - Dashboard

Result

- The average performance score is 85.
- The score falls as years of service increase (92, 85, 78), but there are only 3 employees.
- Each department has one employee.